Ali Abdari

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PhD (2025) in Artificial Intelligence with 6+ years of experience developing and deploying deep learning systems for image and video understanding. Expertise in real-time video segmentation, multi-object tracking, action recognition in raw and compressed video, and multimodal vision-language models. Experienced in adapting, fine-tuning, and profiling large foundation models, including ViT-L-based segmentation models for latency-constrained inference in PyTorch, ONNX, and TensorRT. Proven track record of translating research into production systems in healthcare and automotive, and publishing at ICMR, MMM, IEEE TMM, and ACM TOMM.

Professional Experiences

Computer Vision and AI Researcher — SMC | a DGS Company 2026-Present

- **Real-Time Medical Image/Video Segmentation and Tracking (Hospital of Padua):** Fine-tuned SAM 3 (ViT-L) with LoRA and SegFormer for surgical/clinical footage | Profiled the inference pipeline stage-by-stage and accelerated it 4× with ONNX and TensorRT | Integrated the models into CVAT to have semi-automate annotation
- **3D Deep Learning for Automotive (Stellantis):** Evaluating B-rep, mesh, and point-cloud DL models for automotive geometry tasks | Benchmarking representation trade-offs for accuracy and inference cost
- **LLM-Based Report Generation:** Built an agentic system that answers natural-language questions over company databases and generates reports; deployed to multiple clients
- **Warehouse Layout Optimization:** Using simulated annealing and Genetic algorithms to minimize item retrieval cost

Computer Vision Engineer — Caica 2022-2023

- **Primate Behavior Analysis via Video** (German Primate Center, DPZ)**:** Developed a system for detection, tracking, keypoint detection, and behavioral analysis | Fine-tuned YOLO models | Used DeepSORT for multi-object tracking
- **Fatigue Detection System Using Facial Keypoints:** Developed a fatigue detection system using MediaPipe to detect facial keypoints and measure eye blinks | Applied to factory workers to improve workplace safety
- **AI-Based Assistive System for Patients Using Eye Tracking:** Developed an assistive system using MediaPipe to track eye direction and detect eye blinks | Enabled users to interact with virtual keyboards and type using eye movements

Computer Vision Engineer — Iran's National Elites Foundation 2019-2021

- **Traffic and Pedestrian Video Synopsis:** Analyzed movements in urban and non-urban areas using DeepSORT and FairMOT | Detected abnormal traffic patterns | Generated concise summaries of long videos by temporally rearranging moving-object trajectories | Reduced long video sizes to approximately 1/5 of their original size to facilitate efficient review by users

Android and Python Developer — Mosbat Company 2018-2020

Python Developer — Vestaak Company 2017-2018

Education

Ph.D. in Artificial Intelligence – University of Naples Federico II, University of Udine, Italy 2023-2025
Area: Complex 3D Scenes and Metaverses Retrieval using Natural language

M.S. in Computer Science – Kharazmi University, Iran 2016-2019
Area: Action Recognition in compressed video

B.S. in Computer Engineering – Kharazmi University, Iran 2012-2016

Skills

Architectures — Vision Transformers (ViT, ViTDet), SAM 3 / SAM-family, DETR-style detectors, SegFormer, CLIP-style contrastive vision-language models, YOLO, LoRA fine-tuning, 3D point-cloud encoders

Optimization & Deployment — ONNX, TensorRT, inference profiling, FlashAttention, torch.compile, mixed precision, CVAT, Docker, Kubernetes, AWS, Azure ML

ML/DL Frameworks — PyTorch, TensorFlow/Keras, HuggingFace, OpenCV, Dlib, MediaPipe, NumPy, Pandas, MatPlotLib, SciKit-Learn

Coding — Python, C++, C, MATLAB, SQL

Languages — English (C1, 7.5 IELTS band score), Italian (C1), Persian (Native)

Academic Activities and Experiences

Organization — CV4Metaverse Workshop at ECCV 2024, Italy and CVPR 2025, USA 2024,2025

Supervision — a Ph.D. student and a research fellow on ML for agricultural and genomic data 2024-2025

Organization — Metaverse Apartment Retrieval Challenge at CV4Metaverse at ECCV 2024, Italy 2024

Teaching Assistant — Deep Learning, University of Udine 2023-2024

Computer Vision and Machine Learning Researcher — Kharazmi University 2021-2023

Teaching Assistant — Seven C.S. courses for the B.S. and M.S. degrees, Kharazmi University 2013-2019

Selected Publications (Google Scholar)

1. **Abdari, A.**, Falcon, A., & Serra, G. (2026). Retrieving Relevant Metaverses Using Hierarchical Features. ACM Transactions on Multimedia Computing, Communications and Applications.
2. Falcon, A., **Abdari, A.**, & Serra, G. (2025). ALCER3D: Adaptive Learning Constraints for Enhanced Retrieval of Complex Indoor 3D Scenarios. IEEE Transactions on Multimedia.
3. **Abdari, A.**, Falcon, A., & Serra, G. (2024). A language-based solution to enable metaverse retrieval. In International conference on multimedia modeling. Cham: Springer Nature Switzerland.
4. **Abdari, A.**, Falcon, A., & Serra, G. (2024). Adoctera: adaptive optimization constraints for improved text-guided retrieval of apartments. In Proceedings of the 2024 International Conference on Multimedia Retrieval.
5. Honarjoo, N., **Abdari, A.**, & Mansouri, A. (2024). Violence detection in compressed video. Multimedia Tools and Applications.